

Dielectric relaxation and ionic transport in poly(ethylene carbonate)-based electrolytes

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To study the ion-conductive and dielectric properties of polymer electrolytes based on poly(ethylene carbonate) (PEC) with lithium bis(trifluoromethanesulfonyl)imide (LiTFSI), the complex permittivity and conductivity were measured using broadband dielectric spectroscopy. The temperature dependence of the relaxation frequency and ionic conductivity for PEC-LiTFSI electrolytes (1–200 mol%) indicates that the segmental motion of PEC chains decreases with the addition of just 1 mol% of Li salt and increases with increasing concentration above 10 mol%. According to the Walden rule for PEC-based electrolytes, the value of deviation from the reference line increased, and the fragility and decoupling exponents decreased with increasing salt concentration. These results indicate that there are large numbers of ion pairs and aggregated ions, which imply low ionicity and reduced fragility in highly concentrated PEC-based electrolytes. Copyright © 2016 John Wiley & Sons, Ltd.

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Keywords: broadband dielectric spectroscopy; poly(ethylene carbonate); polymer electrolyte; dielectric relaxation; ionic conductivity

INTRODUCTION

With growing expectations on the next generation of rechargeable batteries, solid polymer electrolytes (SPEs) have recently generated interest as a result of their nonflammability, flexibility, and light weight.^[1–3] In previous decades, there have been many studies of SPEs based on poly(ethylene oxide) (PEO) in which the macromolecular design of polymers,^[4] improvements in salt solubility,^[5] and addition of inorganic fillers^[6,7] were all used to enhance the ionic conductivity. Unfortunately, ionic transport of polyether-based electrolytes is coupled closely to their segmental relaxation, so that these approaches may never reach the desired conductivity level of 10^{-3} S/cm at ambient temperature.^[8–10] Consequently, we focus on the carbonate structure as a unit for a novel polymer main chain, which can arise by the alternating copolymerization of CO₂ and epoxides.^[11–15] We have long studied the ion-conductive behavior of specific poly(ethylene carbonate) (PEC)-based electrolytes. Li transference numbers (t_{Li+}) for PEC-based electrolytes are high ($t_{Li+} > 0.5$),^[13] unlike typical PEO-based electrolytes ($t_{Li+} < 0.3$).^[16,17] Furthermore, PEC can dissolve a large amount of Li salt,^[12] resulting in a polymer-in-salt system in which the salt is the dominant component discussed originally by Angell *et al.*^[18] This is applicable as a super-concentrated electrolyte system and has recently received attention for its outstanding electrochemical properties.^[19,20] As a result, PEC is a candidate for the base polymer in the next generation of SPEs. In our recent study,^[21] the dependence on the salt concentration of the dielectric relaxation and ionic conductivity for a novel polymer electrolyte consisting of PEC and lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) was analyzed using broadband dielectric spectroscopy. This is a powerful technique for studying the ion-conductive behavior of SPEs.^[21–26] The mechanism of ionic transport in PEC-based electrolytes is still not understood; however, if that mechanism can be

determined in detail, it is expected to facilitate application in novel “high-concentrated polymer electrolytes” and has promise in the design of novel polymers, improvement of salt properties, and selection of suitable additives for SPEs. In the present study, we report detailed studies of the temperature dependence of the ionic conductivity and dielectric relaxation for PEC-LiTFSI electrolytes over a broad range of salt concentrations.

EXPERIMENTAL

Poly(ethylene carbonate) ($M_n = 100,000$, $M_w/M_n = 2.7$ from GPC, Empower Materials Inc., New Castle, DE, USA) and lithium bis(trifluoromethanesulfonyl) imide (LiTFSI, Battery Grade, Kishida Chemical Co., Osaka, Japan) were used for the preparation of the electrolytes. A chloroform solution of PEC was dripped into excess methanol, and the precipitated PEC was then dried in a vacuum oven at 60 °C for 24 h. An acetonitrile solution of PEC and LiTFSI at a concentration in the ratio of one Li⁺ to between 0 and 200 mol% monomer units ($[Li^+]/[monomer] \times 100$) was cast onto a plastic petri dish. The solution was dried in N₂ at 60 °C for 6 h and dried under vacuum at 60 °C for

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at least 24 h. After the drying process, the sample was introduced into an Ar gas-filled glove box (H_2O , $\text{O}_2 < 1$ ppm). The sample for measurement was sandwiched between 6 mm ϕ stainless steel electrodes, with a 120- μm -thick silica glass fiber as spacer. The cell was placed under a pressure at 100 kPa and a temperature at 80 °C for 1 h so as to improve interfacial contact between the electrode and electrolyte and to adjust the sample thickness to the spacer. The cell was finally sealed into the glove box in an Al-laminated bag.

The complex permittivity (ϵ^*) and the complex conductivity (σ^*) were measured in the frequency range 10 to 10 MHz using two instruments: The first is a laboratory-made dielectric spectrometer consisting of a function generator, a charge amplifier, and a digital data acquisition system. The second is an impedance analyzer (4192A, Yokogawa-Hewlett-Packard, Hachioji, Tokyo, Japan). Measurements were performed during step-wise cooling at 10 °C intervals from 100 to −20 °C.

RESULTS AND DISCUSSION

Analysis of dielectric spectra

Figure 1 shows the frequency dependence of the storage permittivity (ϵ') and loss permittivity (ϵ'') spectra for PEC-LiTFSI (10 mol%) in the temperature range −20 to 100 °C. In the ϵ' spectra (a), two clear steps can be observed; one is attributed to dielectric relaxation processes at high frequencies, and the other is more intense and attributed to electrode polarization processes at low frequencies. For the ϵ'' spectra (b), there was a linear increase at the medium frequencies, which is associated with

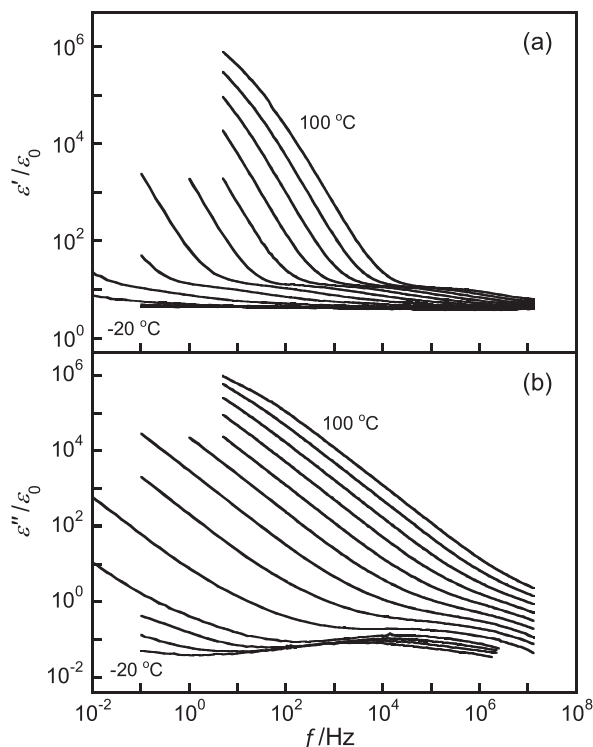


Figure 1. (a) Real and (b) imaginary components of broadband dielectric spectra for poly(ethylene carbonate) with lithium bis(trifluoromethanesulfonyl)imide (10 mol%) at every 10 °C interval from −20 to 100 °C.

DC conduction, and also dielectric loss peaks with weak intensities at the higher frequencies, which are partially hidden by the DC conduction phenomenon. With decreasing temperature, these spectral features shifted to lower frequencies, and the loss peaks became more distinguishable. As in previous studies,^[21–24] quantitative analyses of these spectra involve fitting the observed data with the following empirical function:

$$\epsilon^* = \epsilon_\infty + \sum_k \frac{\Delta\epsilon_k}{\{1 + (i\omega\tau_k)^{\beta_k}\}^{\alpha_k}} + \frac{\sigma_{dc}}{i\omega} \left\{ 1 - \frac{1}{1 + (i\omega\tau_{el})^\gamma} \right\} \quad (1)$$

where ϵ_∞ is the instantaneous permittivity, $\Delta\epsilon_k$ is the relaxation strength, τ_k is the relaxation time, and α_k and β_k are parameters describing the distribution of relaxation times with the k th dielectric relaxation process. Figure 2 shows an example of curve fitting of the ϵ'' spectra for a PEC-LiTFSI (10 mol%) electrolyte at 20 °C. From this figure, two dielectric relaxation processes can be seen at low and high frequencies, which are assigned as α and β , respectively. The α relaxation peak is mostly obscured by the contribution of the DC conductivity (straight line gradient of −1) and has an asymmetric shape (Havriliak–Negami type).^[27] The β relaxation peak is clearly manifest as peaks at low temperature, with symmetric profile (Cole–Cole type).^[28] The observed spectra (open marks) can be clearly divided into these peaks using Equation (1) (solid lines), so that we can evaluate quantitative features as dielectric and conductive parameters.

Temperature dependence

The temperature dependence of the relaxation frequency f_R of the dielectric relaxation events for neat PEC and PEC-LiTFSI electrolytes is shown in Figure 3(a). These dielectric relaxation parameters, α and β , are shown for both neat PEC and PEC-LiTFSI electrolytes. High-concentrate electrolytes (100 and 200 mol%) showed no β relaxation from the dielectric loss peak, because the small β relaxation peak was completely hidden by the α relaxation where the ion-dipole interaction occurs.^[21] These temperature dependences differ from previous systems; the α and β relaxations correspond respectively to Vogel–Tamman–Fulcher type and linear temperature dependence (Arrhenius type). The α relaxation is therefore attributed to the cooperative segmental motion of polymer main chains, and the β relaxation is often identical with the so-called Johari–Goldstein (JG) β -relaxation,^[29] which is observed in various kinds of glass forming materials and

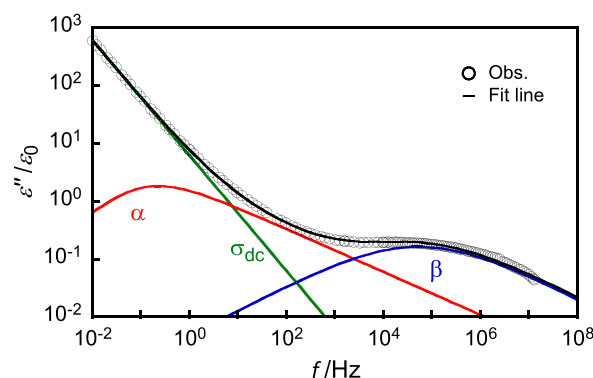


Figure 2. Fitting results of the ϵ'' spectrum of poly(ethylene carbonate) with lithium bis(trifluoromethanesulfonyl)imide (10 mol%) at 20 °C. This figure is available in colour online at wileyonlinelibrary.com/journal/pat

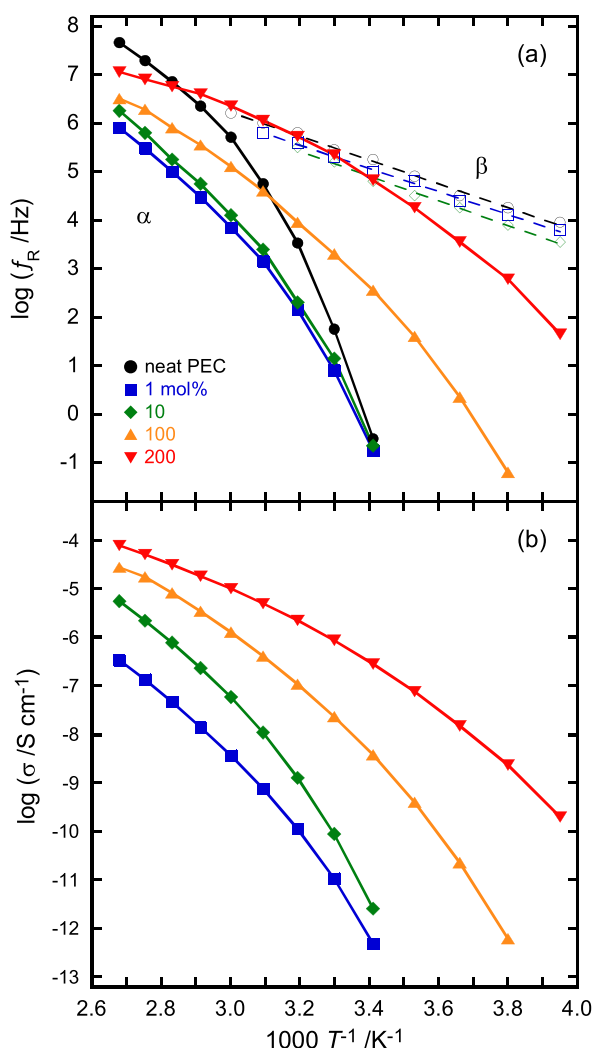


Figure 3. Temperature dependence of (a) relaxation frequencies (filled and open plots indicate α and β relaxation, respectively) and (b) ionic conductivity for neat poly(ethylene carbonate) and poly(ethylene carbonate) with lithium bis(trifluoromethanesulfonyl)imide electrolytes. This figure is available in colour online at wileyonlinelibrary.com/journal/pat

is believed to be an inherent property of glassy matter.^[30,31] From Figure 3(a), the value of f_R for α relaxation for 1 mol% electrolyte shifted to lower frequencies relative to neat PEC but shifted to higher frequencies at higher salt concentrations, above 1 mol%. This behavior is consistent with previous measurements of the dependence on the salt concentration of dielectric properties for a PEC-LiTFSI electrolyte at 40 °C.^[21] We believe that the segmental mobility at first decreases because of ion-dipole interaction but then increases at higher concentrations because of the weakening of intramolecular and intermolecular interactions. Interestingly, the Vogel–Tamman–Fulcher curves for the α relaxation change drastically with increasing salt concentration, and the α relaxation for neat PEC has highest f_R at high temperature. On the other hand, the β relaxation of neat PEC scarcely changes with increasing salt concentration. The complex conductivity ($\sigma^* = \sigma' - i\sigma''$) was obtained from the complex permittivity ($\varepsilon = \varepsilon' - i\varepsilon''$) according to the relation $\sigma^* = i\omega\varepsilon^*$ ($\sigma' = \omega\varepsilon''$, $\sigma'' = \omega\varepsilon'$), where ω is the angular frequency $2\pi f$ (f denotes the measurement frequency). The ionic conductivity σ was obtained from the flat region in the σ' spectra. The

temperature dependence of σ for PEC-LiTFSI electrolytes is shown in Figure 3(b). The value of σ obviously increases with increasing salt concentration. Comparison of Figure 3(a) and (b) clearly reveals the correlation between ionic transport and segmental motion in PEC-based electrolytes. The increase in σ below 10 mol% contributes to the increase in the number of carrier ions, and the temperature dependence of σ and f_α above 10 mol% is very similar in behavior. The temperature dependence of the segmental relaxation frequency (f_α) for PEC-LiTFSI 100 mol% is also shown in Figure 4, with σ data. The data for f_α are in complete agreement with data for σ , which means that ionic transport in PEC is caused by the coupling of segmental (structural) dynamics, in the typical polyether system.^[32]

Walden plot analysis

For further quantitative analysis of dielectric and ion-transport properties in PEC-based electrolytes, we used the Walden plot analysis. This method is inspired by a series of studies performed by Sokolov *et al.*^[8,9,25,33] The Walden rule provides a very useful classification for ionic conductors. When the molar conductivity (Λ) is shown as a function of the fluidity ($1/\eta$) of an electrolyte on a double-logarithmic plot, a straight line is obtained, ideally with a slope of one, according to the Walden rule. The reference line can be divided into two regimes; one is the superionic (above side), and the other is the subionic (below side).^[33] Dilute LiClO_4 aqueous solution^[34] was used as a reference electrolyte in this study. To apply the Walden plot to PEC-based electrolytes, the reciprocal value of the α relaxation time ($1/\tau_\alpha$) can be substituted for the fluidity. Figure 5 shows the correlation between Λ ($\Lambda = \sigma/c$, where c is the molar concentration) and $1/\tau_\alpha$ for PEC-LiTFSI electrolytes. Plots of low concentration electrolytes (1 and 10 mol%) are almost the same as the reference line at higher temperature, but the slope is slightly different from the reference line, and the plots increase with decreasing temperature into the superionic regime near the glass transition temperature (T_g). Plots of high concentration electrolytes (100 and 200 mol%) are

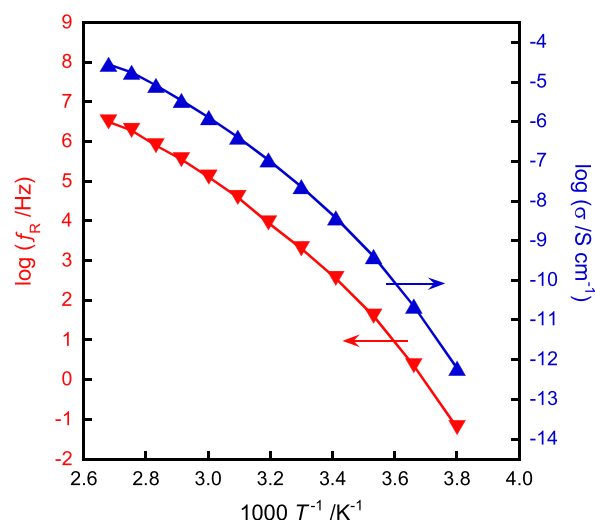


Figure 4. Temperature dependence of the α relaxation frequency (f_α) and ionic conductivity (σ) for a poly(ethylene carbonate) with lithium bis(trifluoromethanesulfonyl)imide electrolyte (100 mol%). This figure is available in colour online at wileyonlinelibrary.com/journal/pat

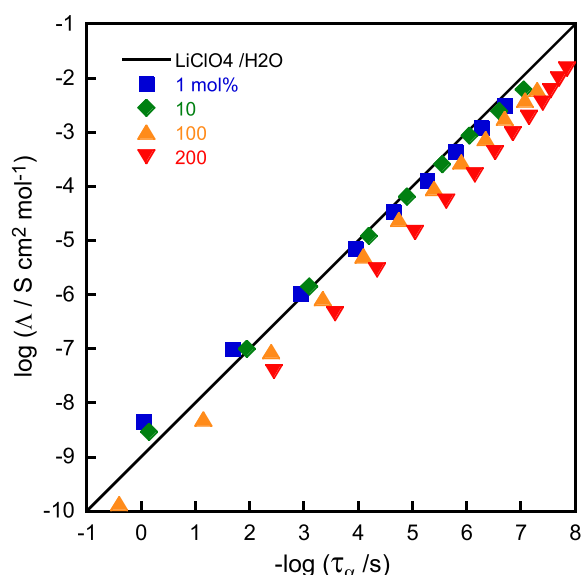


Figure 5. Walden plots of poly(ethylene carbonate) with lithium bis(trifluoromethanesulfonyl)imide electrolytes and LiClO₄ aq. as a reference line.^[8] This figure is available in colour online at wileyonlinelibrary.com/journal/pat

in the subionic regime, and the slope is very close (and parallel) to the reference line. Based on the Walden plots, we propose two characteristic behaviors of PEC-based electrolytes; one is the plot separated from the reference line, which is related to ionicity, and the other is the plot with a gradient of one as the decoupling exponent.

To better understand the behavior shown in Figures 3 and 4, the values of ΔW , the fragility (m), and the decoupling exponent (ε) of neat PEC and PEC-LiTFSI electrolytes were calculated and are summarized in Table 1. We first evaluated the “ionicity,” which is commonly used in measuring the degree of ionic character in ionic liquids.^[35] To determine the ionicity of PEC-based electrolytes, we compared the ionic mobility with the structural relaxation time obtained from the Walden plots.^[36,37] Angell and co-workers determined the ionicity by measuring the deviation from the reference line and denoting the difference as ΔW .^[38] A larger ΔW implies lower ionicity (lower free ion concentration). Every deviation from the reference line was estimated at a fixed value of $\log(\tau_s^{-1}) = -4$. In Table 1, ΔW increases strongly with increasing salt concentration, and the greatest value is at a concentration of 200 mol%. This behavior implies the presence of extensive ion pairing or the formation of aggregated ions in highly concentrated PEC electrolytes (see FT-IR spectra

in Fig. S1 of Supporting Information). Indeed, it has been reported that the ionicity for typical liquid electrolytes decreases with increasing salt concentration, because of the formation of ion-pairs in the concentrated solution.^[39]

Secondly, we determined the “fragility,” which is commonly used to characterize the dynamics of the temperature dependence of supercooled liquids near the glass transition. The value of m quantifies the deviation from the Arrhenius law of the $\log \tau_s$ vs (T_g/T) plot and is determined from $d(\log \tau_s)/d(T_g/T)$ at T_g , where τ_s is the structural relaxation time.^[40] In Table 1, the values of m for neat PEC and PEC-LiTFSI electrolytes are summarized, and a detailed analysis of fragility is shown in Fig. S2 of Supporting Information. For neat PEC, m was very large, whereas the m value of typical ion-conductive polyethers, e.g. PEO ($m \sim 20$),^[41] is relatively low. Recent theoretical and experimental studies^[42–44] have determined that the fragility of polymers depends on the relative rigidity of the chain structure. Polymer chains with rigid backbones suffer from appreciable packing frustration, which results in a loose structure and a strong non-Arrhenius temperature dependence of the segmental dynamics.^[45] Reportedly, PEC has intramolecular interactions between carbonyl oxygens (C=O) and methylene hydrogens (C-H), resulting in the C-C bond forming a *gauche* configuration.^[46] This local binding structure appears to give PEC a relatively loose packing structure. However, the m value of neat PEC decreased drastically to 77 upon the addition of a small amount of LiTFSI (1 mol%). This is because (as described previously^[21]) the local binding structure of neat PEC relaxes with the addition of salt, and dissociation occurs to form ion-dipole interactions; a dense packing structure with enhanced local rotational mobility is likely to form. The m value of highly concentrated electrolytes was further reduced because of the large number of ion-pairs and aggregated ions densely filling the local space.

Finally, we determined the “decoupling exponent” proposed by Sokolov *et al.* As in their report,^[9] we plotted $\sigma T \tau_\alpha$ as a function of τ_α for PEC electrolytes, in Fig. S3 of Supporting Information. The decoupling behavior near T_g can be estimated from a power law relation ($\sigma T \tau_\alpha \sim \tau_\alpha^\varepsilon$), which reduces to the classical Stokes–Einstein relation for $\varepsilon = 0$; the exponent ε can be used to quantify the degree of decoupling.^[25] As seen in Table 1, the value of ε for 10 mol% electrolyte decreased more than twice as much as that of 1 mol% sample, while ε approaches zero with increasing salt concentration, where ionic transport occurs via coupling with the polymer segmental motion. The degree of decoupling appears to be closely related to the fragility of polymer electrolytes.^[25] Similarly, in the present study, the decoupling exponent and the fragility both decreased with increasing salt concentration. The loose structure of neat PEC may deteriorate by dissociation of the Li salt because of the coupling structure and segmental motion involved in ionic transport.

CONCLUSIONS

We have analyzed the dependence on temperature and salt concentration of the dielectric and conductive spectra for PEC-LiTFSI electrolytes, using broadband dielectric spectroscopy. The Arrhenius plots of the relaxation frequency and ionic conductivity gave rise to the following conclusions: 1) the segmental mobility of PEC was reduced upon the addition of a small amount (1 mol%) and dissociation of LiTFSI; 2) the conductivity increased with increasing carrier ions; and 3) the segmental mobility was

Table 1. Values of ΔW , fragility m and decoupling exponent ε of neat PEC and PEC-LiTFSI electrolytes

mol%	ΔW	m	ε
0	—	120	—
1	0.17	77	0.174
10	0.18	75	0.083
100	0.39	63	−0.015
200	0.91	64	−0.003

PEC, poly(ethylene carbonate); PEC-LiTFSI, poly(ethylene carbonate) with lithium bis(trifluoromethanesulfonyl)imide.

enhanced at more than 10 mol% of salt concentration. According to the analysis of the Walden plot for PEC-based electrolytes, the value of ΔW increased and the fragility and decoupling exponent decreased with increasing salt concentration. These observations imply that there are large amount of ion pairs and aggregated ions, which imply low ionicity and reduced fragility ($m \sim 120$ to 63) in the highly concentrated electrolytes, and a coupling ion-transport mechanism can be seen ($\varepsilon \sim 0$). We conclude that these novel PEC-based electrolytes are suitable materials for batteries in the future, maintaining high fragility despite salt dissociation so as to realize the decoupling-type ion-transport in the polymer.

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